

Did you pay the last time you took the MBTA?



Why this Matters :



Revenue Leakage: Widespread free-riding at unmonitored street-level stops directly undermines MBTA's operational funding, reducing resources for service improvements and accessibility initiatives.



Perceived Fairness: Honest riders bear disproportionate costs, while habitual fare evaders exploit low enforcement visibility eroding social norms and perceived justice in public transit.

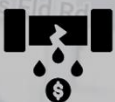


Riders move from intention > action > inaction due to defaults, ambiguity, and low salience of loss.



The Fare Evasion Challenge on the Green Line

Problem Statement:



On the Green Line, there are **78** stops where you can get on without needing to tap your card.



Riders can enter through any door without using a fare gate or inspection.

Evidence:



Qualitative Insight: "I didn't even realize I was supposed to tap the first few times." – BU Student



MBTA staff: "Unless we're checking, most people don't tap. They just ride a few stops and get off."

Secondary Data:



- MBTA estimates it loses as **10 millions+** annually in fare revenue due to unpaid trips.



- In 2022 pilot enforcement on Green Line B branch, compliance rose **30%** temporarily but dropped again post-intervention.



So what?



Revenue Loss

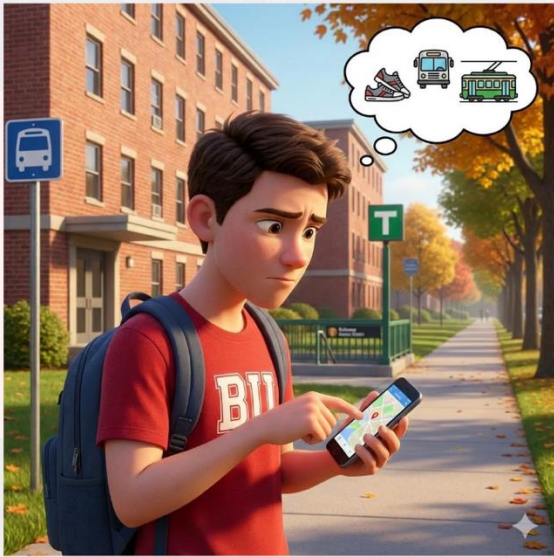


Service Impact



Fairness

Key Rider Journeys



Pre-Ride: Evaluation Moment

- Decide between walking / bussing / Green Line
- Fare payment is not cognitively salient

Boarding: Behavior-Defining Moment

- No physical barrier = board without tapping
- Social norm formed by visible behaviors: "No one taps"
- No signage or cues to activate payment behavior

Ride & Exit: Reinforcement Loop

- No feedback, no reminder, no consequence
- Behavior becomes routinized, forming a long-term habit

Journey Insight: A "required + habitual" journey shaped by social norms, low salience, and poor visual cues — not by intentional dishonesty.

Key Segments that Evade the Most



Rider Segments (Behaviorally Distinct Groups)

Local Students



- Follow what others do
- Board at street-level stops with no barriers
- Don't think skipping fares is a big issue ("it's just one stop")

Young Professionals



- Longer trips are more predictable.
- People are willing to pay more for fair service.
- Fare evasion increases during delays.
- Customers expect consistent service; if not, they justify their frustration.

Financially Vulnerable Riders



- When budgets are tight, tough choices must be made.
- People often feel a sense of scarcity and fear facing penalties.
- We need to create solutions that are fair and don't punish anyone.

Behavioural Root Causes of Non Payment

Fairness-Based Justifications

Delays and bad service change our expectations. Paying for poor service feels like wasting money. People think: "The train is slow today – I shouldn't have to pay for this."



Probability Underweighting

Riders think inspections are uncommon, so the risk seems low. They underestimate the chances of getting caught, making fines feel unreal. In their minds, the likelihood of being penalized is rounded down to nearly zero.

Descriptive Social Norms

Riders notice others getting on without tapping their cards, creating a common behavior. Students are sensitive to these social cues, so they may follow this trend.



Behavioural Root Causes of Non Payment



Choice Architecture Failures



- Street-level stops lack signs or cues
- Confusion about when to tap validation
- Convenience over compliance is the norm

Pain of Paying & Mental Accounting



Tapping for each ride means making many small payments. Monthly pass users tend to tap more since their cost is covered upfront. Students are less likely to tap because each tiny charge adds up and feels painful.

Tap & Win: Overweight Small Rewards



Tap ≥ 15 rides/month = automatic raffle entry

Simple, low-friction rewards system

Prizes: Free monthly pass, \$25-\$50 gift cards, goodies

Prospect Theory, Probability Weighting, Mental Accounting



By setting a clear threshold (15 taps/month), we motivate consistent tapping. The rewards are meaningful but financially manageable for MBTA. It's zero effort for riders—just tap normally & automatically entered.

Gamified Streaks With Peer Visibility



Each day you tap = +1 streak.



Streak badge appears on your MBTA app profile.



Weekly leaderboard for campuses



You can view your friends' streaks (opt-in only)



Inspired by Snapchat streaks, this idea is fun and encourages good habits. Riders can add friends safely, which creates a friendly kind of peer pressure. It promotes social accountability and helps establish routines.

Loyalty Points That Mean Something



Earn 10 points every tap



Redeem for: 500 points = \$5 CharlieCard credit, 1,000 points = 1 free ride and 3,000 points = "Free Week" of unlimited travel



Weekend mornings, late evenings, or rush periods with high evasion get marked "2x Tap Time."



Gives an immediate, tangible benefit from tapping and points create a goal-gradient effect. motivation rises as riders get closer to rewards.

Limited-Edition Charlie Cards



Identity Signaling,
Social Value & Mere
Exposure

- Propose a partnership with Boston museums and sports teams to create limited edition Charlie Cards.
 - Each card showcases unique designs that reflect Boston's artistic and sports culture.
 - Transform the Charlie Card from a standard fare tool into a collectible representing local history.
 - Encourage residents and visitors to view the card as a cherished memento of Boston.
 - Promote the idea of carrying a piece of Boston while traveling around the city.

Descriptive & Injunctive Slogans



Social Norms,
Nudging &
Mere Exposure

Use clear signs about payment rules at stations and on trains. This will help remind people to pay by showing what others expect, to encourage payment by reinforcing social behavior and social expectations

Load Once, Ride Effortlessly



One Transit Wallet in the MBTA app



\$20 / \$40 / \$60 load options



Auto-load triggers when
balance < \$5



Eliminates payment decisions for
each ride, making it easier for users.

Pain of Paying,
Mental Accounting,
Friction Reduction



LOAD ONCE, RIDE EFFORTLESSLY

One Transit Wallet in the MBTA app.

One Transit Wallet
in the MBTA app.



This makes paying easier for riders. By paying in chunks, they think of transit as one cost, which encourages regular use without needing reminders. It simplifies budgeting into one category, "my transit money," and helps create a smooth habit of tapping without having to decide each time.



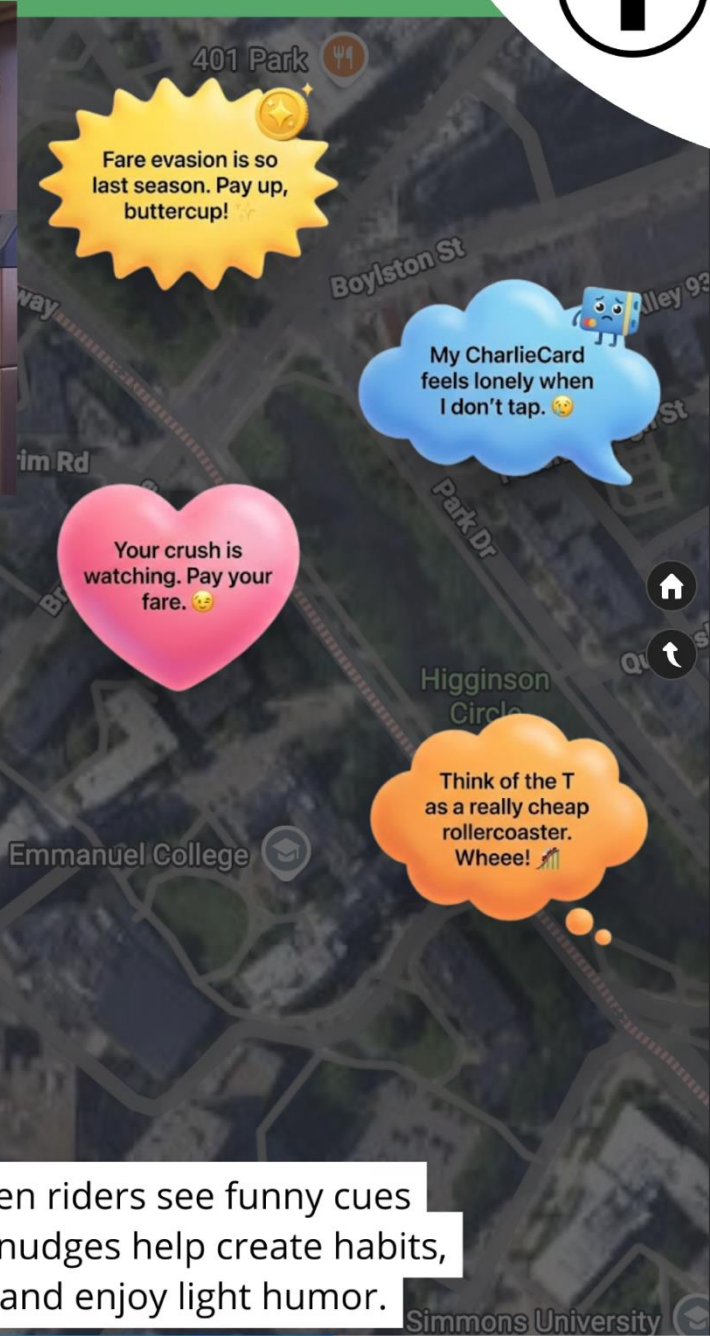
Creative Messaging That Nudges Riders to Pay



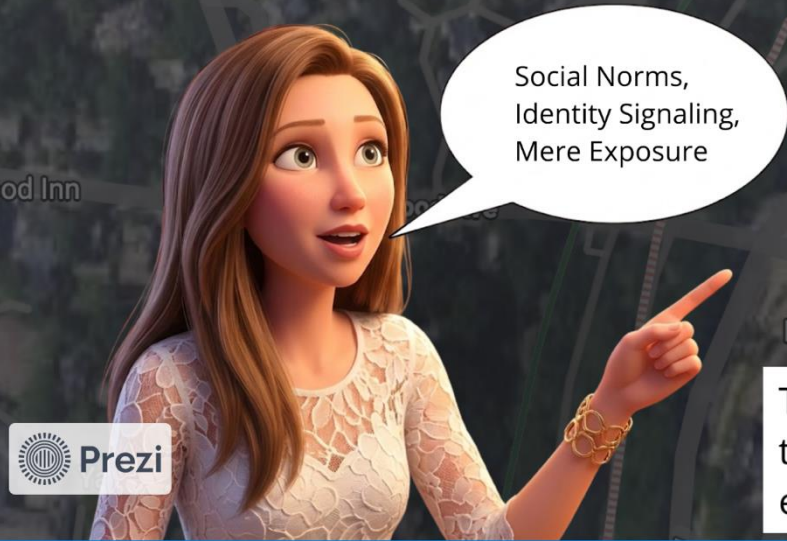
Placement-based messages: Short, humorous nudges positioned at high-visibility touchpoints.



Fun, light in-train announcements to reinforce norms



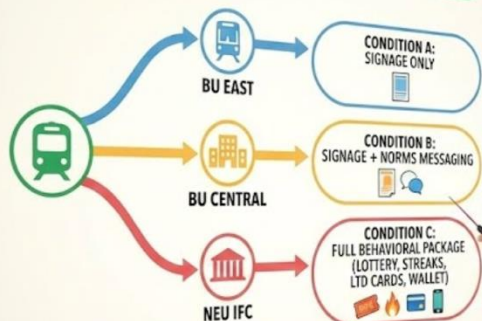
These fun messages make paying feel easy and normal. When riders see funny cues that resonate with them, they find it simple to pay. Friendly nudges help create habits, especially for students who react to their friends' behaviors and enjoy light humor.



Implementation & Measurement



PILOT SCOPE: 6-WEEK PILOT AT 3 STUDENT STOPS



Pilot Scope

- Run a 6-week pilot at 3 high-traffic student stops (BU East, BU Central, NEU IFC).
- Test three conditions:
 - Signage only
 - Signage + Norms Messaging
 - Full Behavioral Package (lottery + streaks + limited-edition cards + wallet)

IMPLEMENTATION & MEASUREMENT

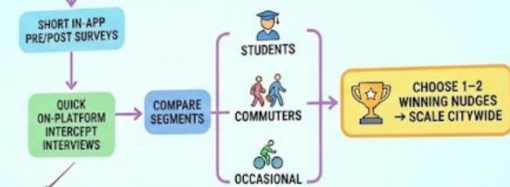
KEY METRICS



KPI's to Focus

- Tap Rate: % of riders tapping per boarding.
- Fare Revenue / Boarding: Compare pilot stops vs control.
- Engagement: # streaks started, # limited cards activated, wallet uptake.

LEARNING PLAN



Learning Plan

- Short in-app pre/post surveys.
- Quick on-platform intercept interviews.
- Compare segments: students vs commuters vs occasional riders.
- Choose 1-2 winning nudges to scale citywide.

We run a small pilot to save money and minimize risk. We have three groups to see which nudges work best for changing behavior. Our main focus is on tap rate, with revenue showing financial impact and sentiment tracking public opinion. This plan helps MBTA expand the best strategies citywide.

 Prezi